

Packets Containing Waveforms

Item Name	Type	Description
VERSION	U3	CCSDS version number, fixed 0b000
TYPE	U1	CCSDS type, fixed 0b0 for telemetry
SEC_HDR_FLG	U1	CCSDS secondary header flag, fixed 0b1 for presence of secondary header
PKT_APID	U11	CCSDS Application Process Identifier (APID), either 0x590 or 0x591
SEQ_FLGS	U2	CCSDS grouping flags, fixed 0xb11 for standalone packet
SRC_SEQ_CTR	U14	CCSDS source sequence counter, increments per packet (separate counters per APID)
PKT_LEN	U16	CCSDS packet length, excludes UMS and CCSDS primary header (minus one)
SHCOARSE	U32	Time of packet generation (not data acquisition), as integer seconds since epoch
SHFINE	U16	Time of packet generation, each DN represents 20usec within current second
SCI0AID	U32	Accountability Identifier for this event
SCI0TYPE	U8	Packet data content type: 1: FPGA pre-pended metadata header 2: TOF HG. 4: TOF LG. 8: TOF MG. 10: Target LG (Q_T) CSA 20: Target HG (Q_T) CSA 40: Ion grid (Q_i) CSA
SCI0CONT	U8	Channels being downlinked (same decomposition as above)
SCI0SPARE1	U13	Spare
SCI0PACK	U1	Always 0b1, showing data is bit-packed
SCI0FRAG	U1	If 0b1, then this channel data spans multiple packets and this is not the last packet (use SCI0FRAGOFF to reconstruct)
SCI0COMP	U1	Data is compressed if 0b1, otherwise not
SCI0EVTNUM	U16	Event number for this event
SCI0CAT	U8	Category assigned to this event (for most recent processing operation)
SCI0QUAL	U8	Quality factor assigned to this category
SCI0FRAGOFF	U16	Starting offset for this data when reconstructing data that spans multiple packets, in 32-bit “dwords” not bytes. Either 0x000 or multiple of 0x3F0
SCI0VER	U16	Science CSC Version number for this header
SCI0TIMECOARSE	U16	Lowest 16 bits of integer seconds since epoch for time of this event
SCI0TIMEFINE	U16	Sub-second time of first event, each DN represents 20usec within current second
SCI0SPARE2	U32	Spare
SCI0SPARE3	U32	Spare
SCI0SPARE4	U32	Spare
SCI0RAW	Varies	Padded with zeroes for 32-bit alignment, if needed
SYNCSCI0PKT	U16	Synchronization marker, fixed 0x3333
CRCSCI0PKT	U16	CRC CCITT-16